

Modern AI models of Text

Embedding, Transformer, Training and Shrinking



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NLP

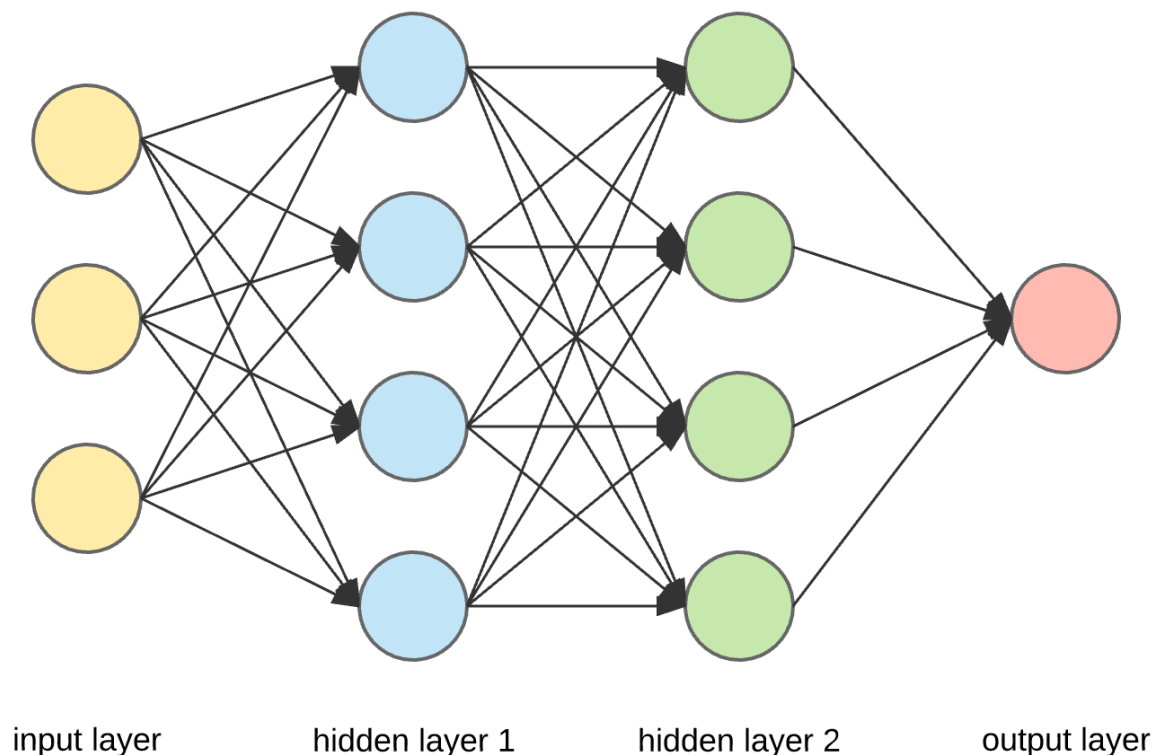
AI adoption





Agenda

- Assumption for this session:
Participants all know what a neural network is. We will refresh rapidly
- Content
 - How do we input Text into any AI model?
 - What is a transformer?
 - Encoders, Decoders, Encoder-Decoder Combinations
 - How do we train these transformers?



$$[4 \times 3] [3 \times 1] = [4 \times 1]$$

$$[4 \times 4] [4 \times 1] = [4 \times 1]$$

$$[1 \times 4] [4 \times 1] = [1 \times 1]$$

A non-linearity between multiplications



Inputs (Embeddings)

The major shift is from treating text as a bag of words to a semantically meaningful, context dependent artifact.



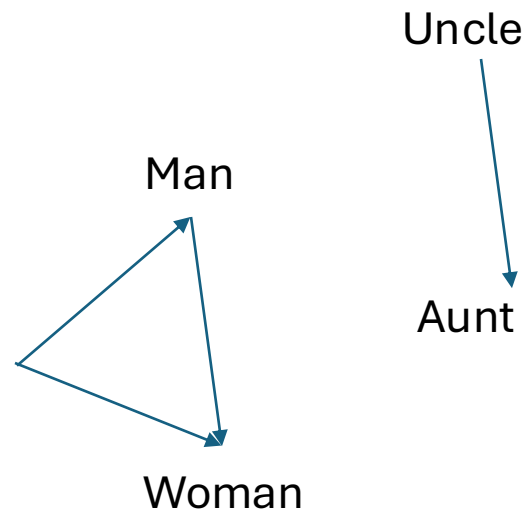
Tokenizing a text

We need to convert text into numbers that neural networks can manipulate.

- First text is automatically converted to bits and bytes ([Convert Text to Binary Code \(Base2 Character & Word Encoder\) - B64Encode](#)) in a computer.
- Greedily combine the most frequent adjacent pairs and proceed to build combinations. This is called tokenization. Token can be a word, word and punctuation, punctuation alone or part of a word. Tokenizers are case, space sensitive.
- Each large model has its own tokenizer. ChatGPT 4.0 uses Tiktokens ([Tokenizer - OpenAI API](#)) and it converted English into 100,277 unique tokens.

Vectorizing a tokenizer

- Tokens are arbitrarily assigned integers. We need to convert them to meaningful representations. We can use one-hot encoding. But, this is sparse and not meaningful.
- [Google Word2Vec](#) showed how to represent words/tokens as vectors of arbitrarily high dimensions. Each dimension can be thought of as a separate concept (independent)
 - All similar words group in one location
 - Directions capture the relationships
 - Uncle – Man + Woman = Aunt
- Tokens are initialized into high dimensional vectors and learned. According to its own admission, ChatGPT uses a pre-learned 12,288 dimension vector for each token.



The key insight is that in an **N-dimensional space**, the number of unit vectors that can be packed while keeping their angles close to orthogonality is:

$$\text{max vectors} \approx e^{cN}$$

where c is a constant (empirically around 0.5–1.0 for many cases). This means:

$$\text{Max near-orthogonal vectors} \approx e^{0.5 \times 12288}$$





Context and Position Embedding

- The context length is how many such tokens can be taken in one prompt (128K for latest ChatGPT). We pad where needed if the length is smaller
- Modern day models are parallelized. Which means, we will lose the crucial sequence information during processing. Hence, we add the position information explicitly.
- These position embeddings are added to the token embeddings for complete packaging of information and then layer normalized.

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$
$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

where

- pos = token position (0, 1, 2, ...)
- i = dimension index
- d_{model} = embedding dimension

Working Example

Assume

- $d_{model} = 4$
- Positions $pos = 0, 1, 2$

pos	PE(0) $\sin(0/10000^0)$	PE(1) $\cos(0/10000^0)$	PE(2) $\sin(0/10000^{2/4})$	PE(3) $\cos(0/10000^{2/4})$
0	0.0000	1.0000	0.0000	1.0000
1	0.8415	0.5403	0.0001	1.0000
2	0.9093	-0.4161	0.0002	1.0000



Architecture

Major shift is understanding that conversation is a combination of context, and knowledge and style. Create blocks that address these

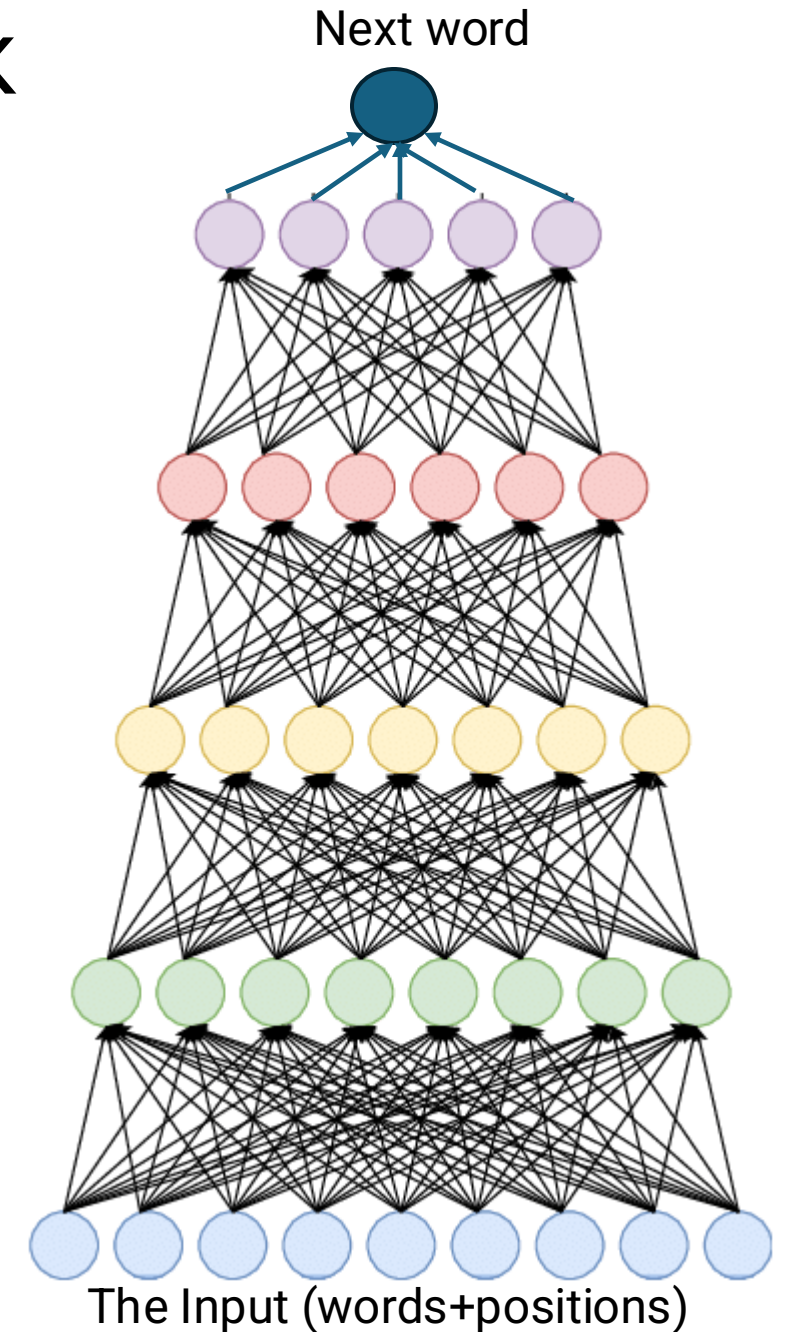


How is conversation built in LLMs

- They start with predicting the next word given a context. This is the unit operation in any text generation. This is called language modelling.
What does predicting the next word take?
 - Ability to understand the context
 - This is a glass of orange _____ (n-1 and n-3)
 - Ability to store and use the knowledge
 - Total century count of Kohli in one-days is _____ (n-2, n-4, n-6, n-7) and still this completion needs some look up knowledge

What about a large neural network

- Neural network is great for storing knowledge
- However, context
 - This is a **glass** of **orange** _____ (n-1 and n-3)
 - The **clouds** are in the _____ (n-4)
 - Total **century count** of **Kohli** in **one-days** is _____ (n-2, n-4, n-6, n-7) and still this completion needs some look up knowledge





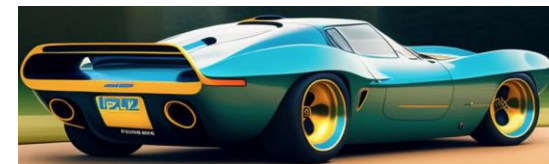
Attention block is the context finder

A separate intervention called “**attention**” handles this problem elegantly. Attention updates the vector of each word based on the context.

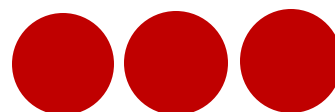
Every word is updated by every other word.

From the embeddings, we initialize query, key and value vectors. Then, $(q.k^T)$ is used to update token vector

After analysing a lot of data, the model learns to update embeddings such that appropriate words influence the word a lot and irrelevant words influence it a little.



Stylish



Blue

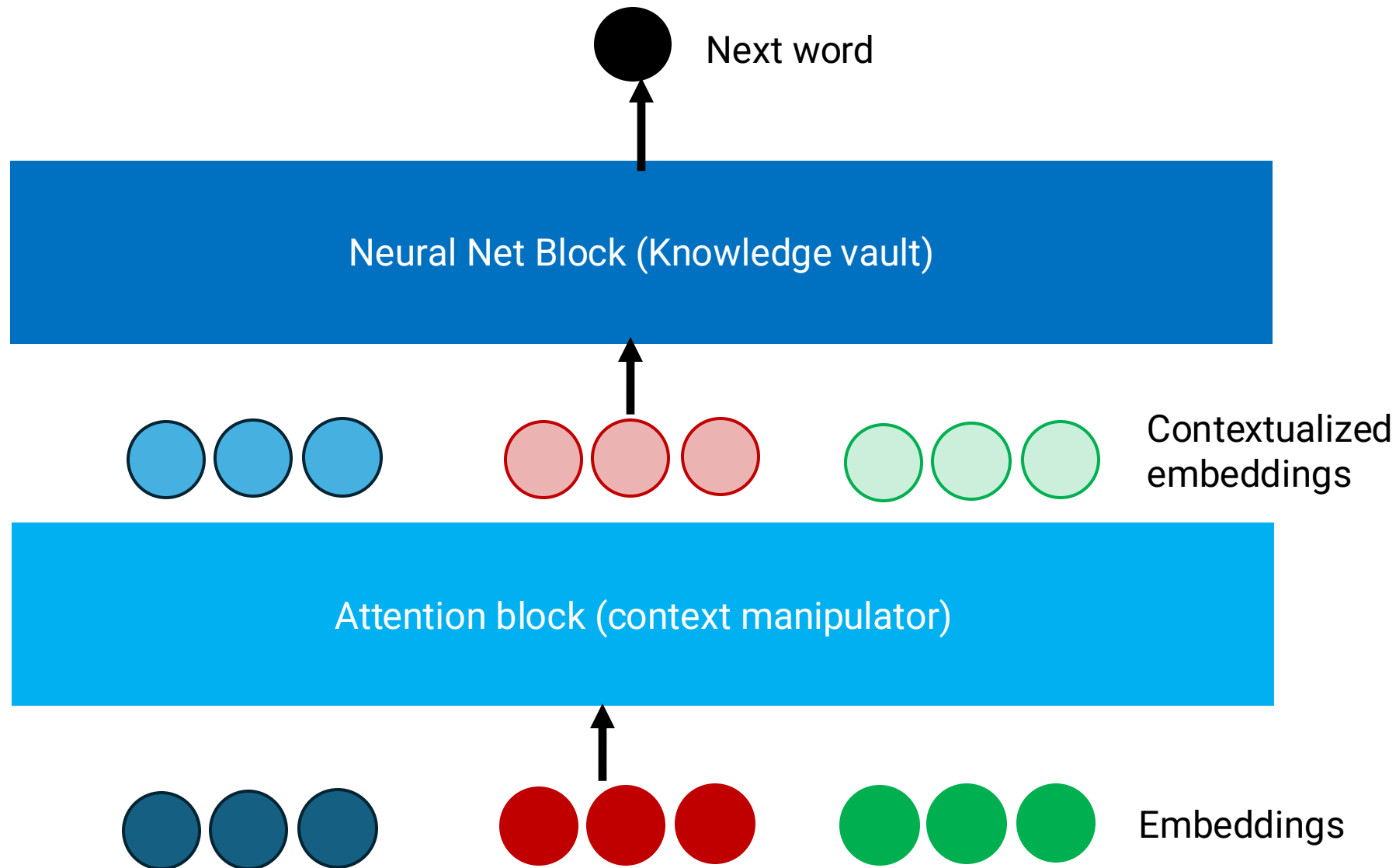


Car





A Transformer architecture



FFN typically stores knowledge

Attention has the ability to manipulate the context.

This combination is called a transformer.

Attention is all you need

An apt title could be “Attention is 30% of all you need!!!” as NNs have 70% weights

Encoder only: Good for classification (BERT)

Decoder only (without cross attention): Generation (GPT)

Encoder + Decoder: Translation

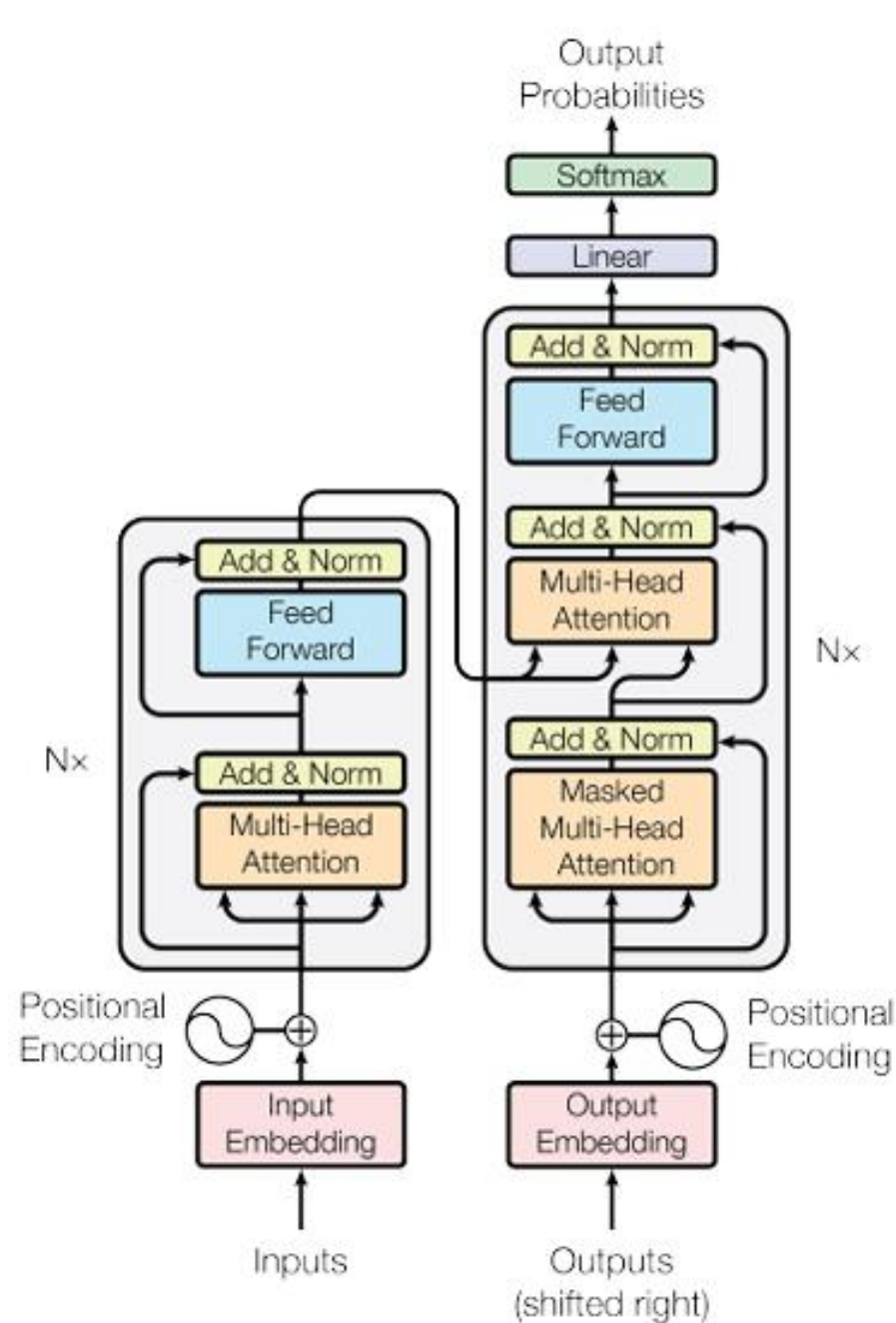


Figure 1: The Transformer - model architecture.

Gen AI (specifically LLM) is humongous DL



TokenIDs (1×4096)



Embedding: $(4096 \times 50257) \cdot (50257 \times 12288) \rightarrow (4096 \times 12288)$



QKV projection: $(4096 \times 12288) \cdot (12288 \times 36864) \rightarrow (4096 \times 36864)$



Split $\rightarrow$ Q(4096×12288), K(4096×12288), V(4096×12288)



Attention: $(4096 \times 12288) \cdot (12288 \times 4096) \rightarrow (4096 \times 4096)$



Weights $\times$ V: $(4096 \times 4096) \cdot (4096 \times 12288) \rightarrow (4096 \times 12288)$



ChatGPT 3 (175 Billion Parameters)

Output proj: $(4096 \times 12288) \cdot (12288 \times 12288) \rightarrow (4096 \times 12288)$



FFN expand: $(4096 \times 12288) \cdot (12288 \times 49152) \rightarrow (4096 \times 49152)$



FFN contract: $(4096 \times 49152) \cdot (49152 \times 12288) \rightarrow (4096 \times 12288)$



Final row: (1×12288)



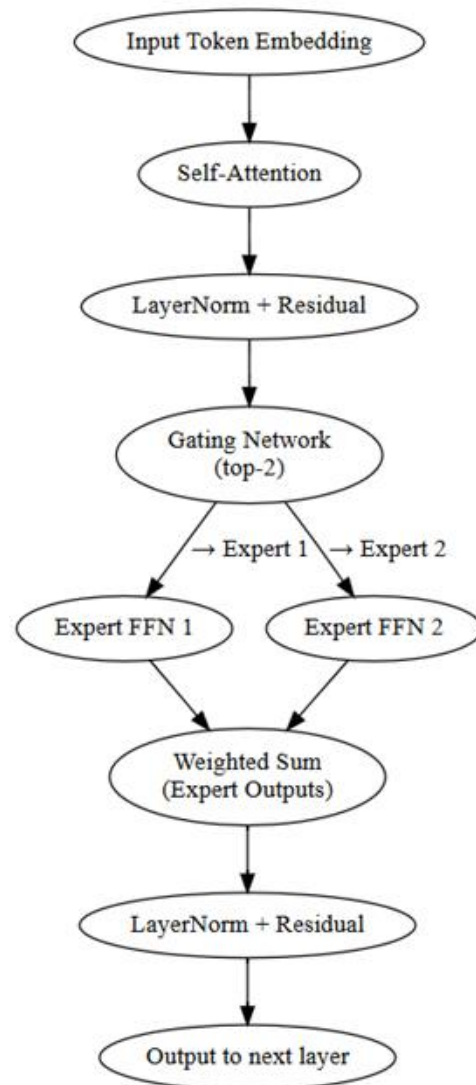
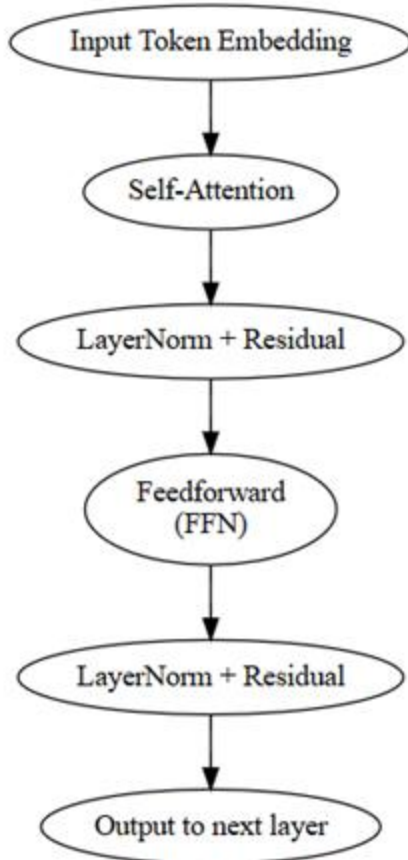
Vocab projection: $(1 \times 12288) \cdot (12288 \times 50257) \rightarrow (1 \times 50257)$



Softmax $\rightarrow$ next-token distribution



Transformers and Mixture of Experts





Training

Supervised, Unsupervised and Reinforcement learning

A training pipeline that combines context, knowledge and style



Pre-Training

- Well curated datasets (fineweb)
 - [HuggingFaceFW/fineweb · Datasets at Hugging Face](#)
- Predict the next token given some previous tokens
 - This is called base model
 - Demo



Supervised learning

- What does generating a whole conversation take (Ability to mimic human style)
- Make people write prompts and answers and train (OpenAI paper)
- Demo of Instruct models



Limitations of Pre-Training and SFT

- SFT lacks explicit negative signals. When a behavior is absent from the training data, the model gradually infers its undesirability through omission. This indirect signal is weak and inefficient. This can lead to hallucinations
- Pre-training and SFT do not provide an effective way to steer a model away from bad outputs.
 - The pre-training dataset contains some data that is unkind, so a model trained with next-token-prediction may sometimes be unkind to the user. How do we explicitly tell the model to be kind to the user?
- They are not also appropriate for specifying abstract goals.
 - E.g. “Follow the user’s instructions helpfully and safely.”
 - E.g. “Refuse a user’s command if it is unethical or dangerous.”



Reinforcement Learning

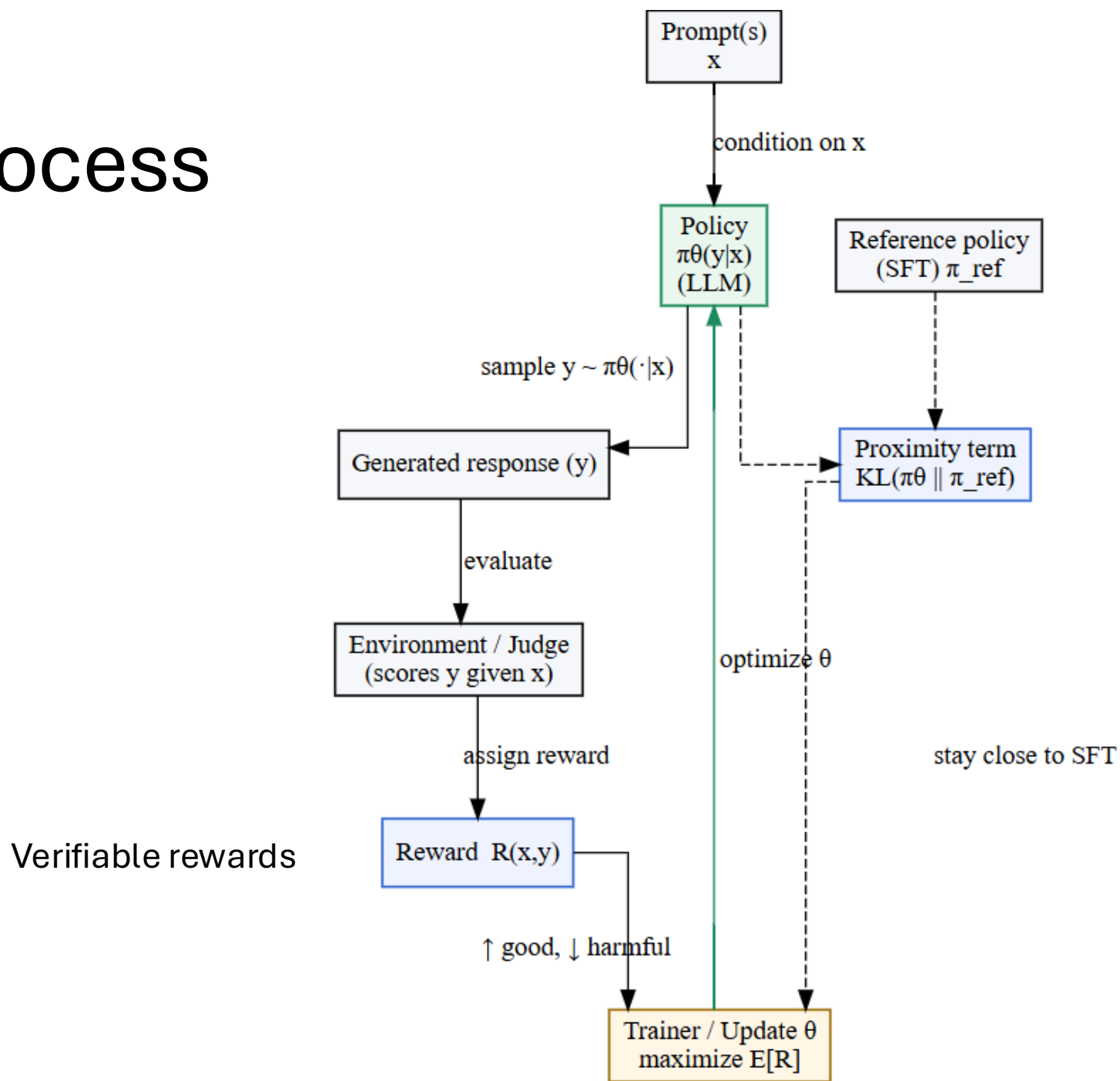
- Instead of copying examples, think of it like training a puppy: you reward good behavior (like sitting on command) and discourage bad ones. However, reward is sometimes difficult to specify as a formal function in conversations.
- RL as a field worked well for games, where reward was winning.
- RL with human feedback (RLHF) uses human feedback to determine the desired behavior, often by training a *reward model*.
 - P. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei, Deep reinforcement learning from human preferences, *NeurIPS*, 2017.
 - B. Ibarz, J. Leike, T. Pohlen, G. Irving, S. Legg, and D. Amodei, Reward learning from human preferences and demonstrations in Atari, *NeurIPS*, 2018.



Why RL works?

- RL provides explicit negative feedback. Actions that lead to lower rewards are penalized. Although there is no credit assignment and all actions leading to a poor outcome are down weighted without fine-grained attribution, the model nonetheless receives direct signals about which behaviors to avoid.
- Studies show RLHF boosts user satisfaction and cuts toxicity (harmful content) by 50% or more (Askell et al., 2021). Without RL, models might be grammatically perfect but unengaging or risky. RL adds that extra polish, like turning a robot into a friendly companion.

RL process





DeepSeek: RLVR

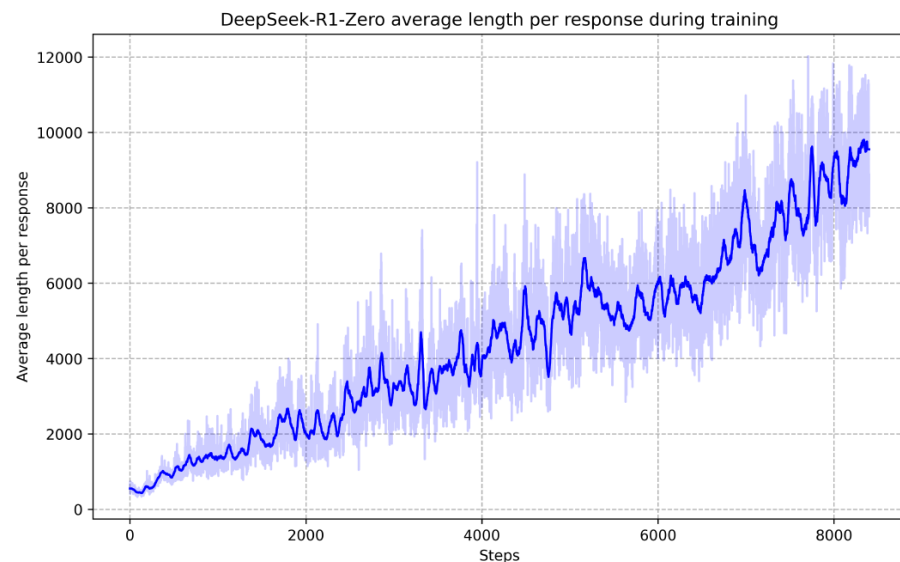


Figure 3 | The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\dots$

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

$\dots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\dots$

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: $\dots$

$\dots$

These are learned by just looking at the correct answers



RLVR and unexpected emergence

- “One of the most remarkable aspects of this self-evolution is the emergence of sophisticated behaviors as the test-time computation increases for models trained with RLVR
 - These behaviors are not explicitly programmed but instead emerge as a result of the model’s interaction with the reinforcement learning environment.”
- These worked for Qwen, Deepseek which were trained on RLVR and not Llama or other models which were not trained!!



RLVR works with cognitive behavior

- Qwen and DeepSeekbase models exhibit certain key cognitive behaviors for reasoning: verification (systematic error-checking), backtracking (abandoning failing approaches), subgoal setting (decomposing problems into manageable steps), backward chaining (reasoning from desired outcomes to initial inputs).

A contrast in behaviors explored by the two models

Verifications
"Let me check my answer ..."

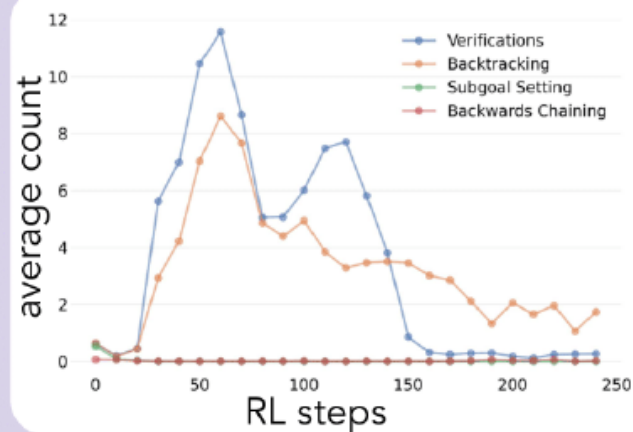
Subgoal Setting
"Let's try to get to a multiple of 10"

Backtracking
"Let's try a different approach, what if we ..."

Backward Chaining
"Working backwards, 24 is 8 times 3"



Qwen



Llama

